Bauer REMI 2 Handset – EWI MIDI Controller

User Manual revision 1.5

References

For a general overview of the REMI project, please refer to the author's website:
http://www.mjbauer.biz/REMI_Intro.htm

Design notes and construction details are posted on the web-page here:
http://www.mjbauer.biz/Build_the_REMI_mk2.htm

The latest firmware release and source code are available for download on GitHub here:
https://github.com/M-J-Bauer/REMI-2-EWI-MIDI-Controller

This edition of the manual is applicable to REMI firmware version 1.5.

Introduction

This manual provides information to help you test and set up your REMI 2 (or compatible) handset when construction is (more or less) complete. It is assumed that the breath pressure sensor module is functioning properly, i.e. the output voltage span of the module is at least 1V, but preferably more, ideally around 3V.

REMI hardware testing and sensor calibration requires the USB "service port" CLI to be functioning. So, let's begin with familiarization of the CLI.

1. Command-Line User Interface (CLI)

The CLI functions like the "Command Prompt" in Windows and the "Terminal" in Linux. Commands are provided to view and set configuration parameters, to view and set preset instrument assignments and for hardware testing, sensor calibration, etc.

To use the CLI, the REMI must be connected to a USB port on your computer. You'll need a USB cable with a "mini-B" or "micro-B" type plug on the REMI end, whichever matches the USB receptacle on your USB/serial bridge (FT232 module).

Terminal Setup

The first step is to download and install a terminal emulator application on your computer. The recommended app for Windows is "PuTTY". Copy the distribution files to a folder on your computer's local drive. To start PuTTY, you just double-click the executable file (putty.exe). It is convenient to create a desktop shortcut.

Connect a USB cable from your computer to the REMI service port (USB-serial connector).

Configure your PC terminal app to use a serial COM port connection. You will need to enter the COM port number that your computer has assigned automatically to the REMI. To find the COM port number for a Windows PC, open the Device Manager utility, accessible from Windows Control Panel. Click 'Ports (COM & LPT)' and identify the USB/Serial bridge used for the REMI. The port number 'COM xx' should be shown.

Configure the baud-rate of your terminal to 38400 baud, with no parity bit.

PuTTY has the 'Backspace' key set to ASCII 'DEL' (127) by default, which is quite useless. Change it to 'Ctrl+H', which is supported by the REMI CLI.

Save the Serial terminal setup, so you don't have to re-configure it again next time.

You should see a start-up message in the terminal window – something like this:

```
* MCU reset *
Bauer {REMI} Handset mk2 – Service Port CLI
Firmware version 1.4.10, Jun 21 2022
>
```

If not, hit the [Enter] key on your computer keyboard. You should see the CLI prompt (**>**).

Command Line Basics

The CLI provides commands for setting up, testing and operating the REMI 2 handset. To see a list of all available commands, enter the command "**lc**" (= list commands) at the prompt as shown below…

```
> lc
cmds        config      default     diag        dump        help
lc          preset      reset       ver         watch
>
```

Another command named "**cmds**" does the same thing (for compatibility with other versions of the CLI used in various other projects).

To get the firmware version installed, type the command "**ver**" as follows:

```
> ver
Firmware version 1.4.10, Jun 21 2022
>
```

A command line consists of a command name followed by a number of "arguments" (*aka* "parameters") separated by one or more spaces. Some commands have no arguments. An argument (*abbr*: "arg") is just a string of characters. There may be some forbidden char's.

A special type of argument is called a "switch". A switch comprises the hyphen symbol (**-**) followed by one or more characters (no spaces). Switches typically follow immediately after the command name. A switch is often used to select a sub-command or a command option.

Command line switches and arguments are generally not case-sensitive.

Most commands have a built-in "help" option which displays a usage summary. To see the usage summary, type the command name followed by the switch **-help**. For convenience, a question mark (**?**) may be typed instead of '-help'. An example follows:

```
> preset ?
Usage #1:  preset    | List Preset MIDI Prgm numbers

Usage #2:  preset <pre#> [=] <value>    | Set Preset value
```

```
   where <pre#> is the Preset number (1..8)

Example:   preset 3 = 67   | Set MIDI Prgm 67 for Preset 3
>
```

In the above example, it can be seen that there are two usages of the command "**preset**". The first usage has no arguments following the command name. This usage will list the MIDI program numbers assigned to the 8 REMI presets. The second usage lets you assign a MIDI program number to a specified preset, as shown in the example.

Most commands, when entered with the switch '-help' or '?', will show command usage information. Otherwise, a command which expects at least one argument, when entered without any, should also show command usage.

In usage summaries, square brackets enclose optional parameters, e.g. [arg]. Braces enclose alternative options, e.g. {-x|-y}, where '|' means 'OR'. So the command can include either switch '-x' or '-y', but not both.

The CLI maintains a history of commands entered. To recall a previously entered command, type Ctrl+R (hold down the Ctrl key and press R) repeatedly until the desired command reappears. This feature is somewhat quirky, but more often than not, it works!

The 5 most frequently used CLI commands in the REMI mk2 handset are:

lc/cmds config preset watch diag

The "**config**" command is explained in the section "**Configuration Options**".

The "**preset**" command is explained in the section "**Instrument Presets**".

The "**watch**" and "**diag**" commands are explained in the section "**Hardware Testing and Sensor Calibration**".

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2. Hardware Testing and Sensor Calibration

USB Service Port Test

Connect your PC to the REMI USB port and start the terminal app. The REMI board should now be powered up. If the CLI is working properly, i.e. you can enter commands and see the responses on the terminal screen, then you can skip the remainder of this test.

If the CLI terminal is not responding, first check that the terminal COM port number is correct. Next, check that the serial port TX and RX signals are wired correctly. The PIC UART TX pin (U1TX/RC6) should be wired to the RX input of the USB-serial bridge (RXI).

Hit [Enter] a few times on the terminal (PC) keyboard. The 'TX' and 'RX' LEDs on the USB/serial bridge should flash.

MIDI OUT Transmitter Test

This is an optional test. If your micro-controller board has been assembled and wired up correctly, the MIDI transmitter/driver circuit should function properly. If you suspect that there may be a defect, then come back and perform this test. You will need an oscilloscope.

Solder a red LED (or a 1k resistor) between pins 4 and 5 on a 5-pin DIN plug. Insert the plug into the MIDI OUT socket wired to the REMI micro-controller board. Hook a 'scope probe onto MIDI OUT– (pin 5 on DIN plug). Hook the 'scope ground clip to GND (pin 3).

Connect your PC to the REMI USB port and start the terminal app. The REMI board should now be powered up. Enter the CLI command "**diag -t**". This will cause the MCU to transmit a data byte (0F hex) continuously. If the MIDI transmitter/driver circuit is functioning correctly, you should see a square wave on the 'scope with frequency = 3.125 kHz.

Pressure Sensor Test and Calibration

Enter the command: "**watch**". The terminal screen will display various data on a single line, continuously updated in real-time, as shown below. The update rate is about 5 times per second.

```
> watch
Hit [Esc] to quit.

TouchPads: 0x0000 | Pad0_ADC:  234 | Pressure:  187 | Modul'n:  455
>
```

To capture the data values at any instant, press the [Enter] key on your PC.

To exit the "watch" function, press the [Esc] key.

The Pressure value is the "raw" ADC reading. The "quiescent" reading (without blowing into the sensor) should be under 200, but not zero. While blowing into the mouthpiece at a comfortable maximum pressure for normal playing, hit the [Enter] key to capture the reading. That reading should be at least 200 ADC units higher than the "quiescent" reading.

The difference between the maximum and minimum (quiescent) readings is called the "span". Set the configuration parameter nick-named "**prspan**" to the pressure span value obtained. (See next section for details.) Your breath pressure sensor is then calibrated!

The higher the span value, the harder you will need to blow to get maximum audio volume (or whatever synth effect(s) may be controlled by breath pressure). Try experimenting with the "prspan" parameter value until you are happy with the range of loudness you can achieve.

Depending on the stability of the pressure sensor, it may be necessary to re-calibrate it from time to time. Note that the quiescent (minimum) pressure reading is automatically recalibrated every time the REMI starts up. Only the span might need to be manually recalibrated and this value is not critical… just so long as it's in the right ball-park.

Modulation Sensor Test and Calibration

The "Modulation Pad" is a sensor wired to produce a voltage proportional to the force applied. The firmware reads the signal and converts it to a data value which is transmitted in MIDI modulation messages (Control Change number 01). You can set the time interval between MIDI modulation messages using the CLI. (The default is 30ms.) Most sound synthesizers can

be set up to respond to CC01 (modulation) messages in various ways. For example, it could be set up to control filter resonant frequency, vibrato depth, noise level, oscillator detune, etc.

Enter the command: "**watch**". Observe the "**Modul'n**" value, which is the ADC input reading for the modulation pad (or whatever other type of sensor you may be using). The minimum value, i.e. without pressing the Mod Pad, should be close to zero, typically below 10.

Press the Mod Pad and verify that the ADC reading increases with the force applied. The maximum value should be above 700, typically about 750. If not, check your Mod Pad sensor wiring and the op-amp buffer circuit. Actually, the op-amp buffer is probably redundant. The sensor signal could be fed directly into the ADC input. The main purpose for the buffer is to protect the MCU from excessive voltage, e.g. ESD, or whatever. It also presents a very low impedance signal to the ADC input.

The LM358 op-amp maximum output voltage is about (Vcc – 1.5) volts, i.e. about +3.5V. This explains why the sensor maximum reading is about 750, not 1023 as you might expect because the PIC18F_K22 has a 10-bit ADC. If you bypass the op-amp or substitute an op-amp with "rail-to-rail" output voltage range, the maximum reading obtainable should be 1023.

Set the configuration parameter nick-named "**modmax**" to the maximum ADC reading obtained. (See next section for details.) Your modulation pad sensor is then calibrated!

If your REMI uses a force-sensitive resistor (e.g. FSR-402) to implement the Modulation Pad, there will be a non-linear relationship between the force applied and the voltage across the load resistor, i.e. voltage at the ADC input. This non-linear characteristic is not wanted. Without compensation, the Mod Pad would feel too sensitive at the low end of the force scale.

Don't worry… the REMI has a configuration parameter named "**modband**" which defines a range of sensor readings at the low end of the scale called a "dead band". In this range, the data value sent in MIDI modulation messages will be zero. If the force applied to the sensor is above the dead-band range, then the MIDI modulation data value will increase proportionally. The "**modband**" parameter is the sensor ADC reading at the top end of the dead-band range. The parameter value should be set so that the Mod Pad feels right when controlling a synth effect, for example vibrato depth, filter cut-off frequency, or whatever. A good starting point is to set the parameter value to half of the maximum reading, e.g. about 350.

Touch-pad Test

Enter the command: "**watch**". Observe the "**TouchPads**" value, which is a hexadecimal number of size 16 bits. Each bit represents the state of a touch-pad: 0 = pad not touched; 1 = pad touched. So, if no pad is touched, the value shown should be zero (0x0000). If one or more pads are touched, the value should be non-zero.

If you convert the hex number to binary, you can determine which pad(s) are touched and hence verify that the touch-pad inputs on the PIC MCU are wired correctly. The pad bit allocations are shown below:

```
                |   octave    | ---------- s e m i t o n e ---------- | LH4 |
Touch-pad ->    | OCT+ | OCT- | LH1 | LH2 | LH3 | RH1 | RH2 | RH3 | RH4 | RH5 |
Bit position -> |  9   |  8   |  7  |  6  |  5  |  4  |  3  |  2  |  1  |  0  |
```

When pad RH5 (or LH4, depending on touch-pad layout) is touched, the hex value displayed should be 0x0001. Pad RH4 touched = 0x0002, RH3 = 0x0004, RH2 = 0x0008, RH1 = 0x0010, LH3 = 0x0020, LH2 = 0x0040, LH1 = 0x0080, OCT– = 0x0100, OCT+ = 0x0200.

The 'watch' command also shows the ADC reading for the touch-pad LH4/RH5. The reading should be higher than 200 when the pad is not touched. The reading should drop below 150 when the pad is touched. The threshold value which determines whether a pad is touched or not is normally set to 150. If for some reason any of your touch pad readings fall outside the normal range, the threshold value can be changed using a configuration parameter named "**thres**". (See next section for details.)

Mode Switch

Provision of a 'Mode' switch is optional. If your REMI omits the switch, it is recommended to tie the MCU switch input pin to GND. The purpose of the Mode Switch is to select one of two sets of configuration parameters, so that it is easy to re-configure the REMI to control one or other of your two favourite synths without needing the CLI.

The Mode switch position (logic state) can be checked using the CLI command: "**diag -m**".

---

3. Configuration Options

Configuration options allow you to set up the REMI to work properly with a MIDI sound module, keyboard synthesizer, or a computer with a software synth (VST) application. These may differ in the MIDI message type used to control audio output level (volume). Keyboard synthesizers typically use "Note-on Velocity", "Aftertouch" (key pressure), or Control Change commands "Channel Volume" or "Expression" to control audio level.

The REMI firmware maintains a bunch of user-settable "configuration parameters" allowing it to function properly with a variety of MIDI-controlled sound modules and synthesizers, whether hardware or software. Some parameters define the physical properties of the handset, for example, the touch-pad layout and fingering scheme. Other parameters are provided for calibration of the breath sensor and modulation pad.

Configuration parameters are stored in flash memory, so their values will be preserved during power-down. Be aware that after a firmware update, however, all parameters will revert to default values, so it will be necessary to re-enter any different values required for your particular REMI and synth setup.

Duplicate sets of synth configuration parameters are provided so that it is quick and easy to switch between two of your favourite synthesizers or sound modules. The required synth parameter set is selected by the 'Mode' switch (if fitted). The '**config'** command addresses only the currently selected parameter set. The default parameter values in Mode 0 (i.e. switch input in low state) are intended for general MIDI synths, whereas Mode 1 defaults are intended for the REMI mk2 Sound Module (*aka* "Bauer EWI Synth").

To view a list of the configuration parameters, enter the command "**config**". This will display the nick-name for each parameter along with its current setting and a brief description, as shown in the following printout.

```
> config
------- Mode 0 Parameters --------------------
chan       1  Midi Basic Channel (1..16)
sysxen     0  Midi Sys.Excl.Msg Enabled (0,1)
prchen     0  Midi Prog. Change Enabled (0,1)
expncc     2  Midi Expression CC number (2,7,11)
expnint    5  Midi Expression Interval (5..50 ms)
expn14b    0  Send 14-bit Expression data (0,1)
legato     1  Legato Mode Enabled (0,1)
velsen     0  Velocity Sense Enabled (0,1)
moden      1  Modulation Pad Enabled (0,1)
pben       0  Pitch Bend Enabled (0,1)
------- Handset Parameters --------------------
fingsc     0  Fingering Scheme (0,1,2,3)
pitch      0  Pitch Offset (0..+/-24 semitones)
thres    150  Touch Sense Threshold (max.250)
prspan   750  Pressure Sensor Span (max.750)
pbspan  1000  Pitch Bend Span (max.1000)
modmax   750  Modulation Maximum (max.1000)
modband  350  Modulation Dead-band (max.500)
```

To get more information about the fingering scheme and pitch setting, enter the "config" command with a plus sign (+) following, as shown below:

```
> config +
Fingering scheme: REMI standard -- pad RH5 flattens
Pitch Offset:    0 semitones    Tenor (C4)
>
```

To change the value of a parameter, enter the command:

config <*param*> [=] <*value*>

… where <*param*> is a parameter nick-name (as in the listing above) and <*value*> is the new value to be set. The square brackets indicate that the equals sign (**=**) is optional.

Example: To set a value of 590 for the pressure sensor span, enter the command:

```
> config prspan = 590    … or …

> config prspan 590
```

Configuration Parameters

Following is a description of the config. parameters implemented in firmware version 1.5.

The first group determines how your synth will respond to MIDI control messages sent by the REMI. There are two sets of parameters in this group, one for each position of the 'Mode' switch, allowing you to swap easily between two favourite synth's.

chan MIDI basic channel number (1..16). The REMI will send MIDI messages addressed to this channel. The channel number must match the MIDI channel set on your synth. The default setting is 1.

sysxen MIDI "System Exclusive" messages enabled. This parameter is intended for use with the REMI mk2 Sound Module. If you are using a REMI Synth, set this parameter to 1; otherwise set it to 0.

prchen MIDI Program Change enabled. Set this parameter to 1 if you want to select synth instrument voices using the PRESET button on the REMI. Otherwise, set it to 0. [Note: MIDI Program Change messages are ignored by the REMI mk2 Synth. "Sys. Ex." message is used instead.]

expncc MIDI Expression Control Change number (1..31). If your synth supports Breath Pressure CC messages, set this param to 2 (default); otherwise, if your synth supports Channel Volume CC messages, set this param to 7; otherwise, set it to 11 for Expression. (See note 1 below.)

expnint MIDI Expression message interval (5..50 ms). This param sets the time interval between transmission of MIDI Expression CC messages. This should be set as low as your MIDI network data flow will permit. (Default: 5ms)

expn14b MIDI Expression data size. MIDI Control Change messages can support 7-bit or 14-bit resolution for the expression data value. If your synth supports 14-bit CC data, set this param to 1; otherwise set it to 0.

The same parameter expn14b also determines the resolution of MIDI Modulation message data (CC01, CC33), i.e. 7 bits or 14 bits (resp.).

legato Legato Mode enabled. If your synth supports legato pitch changes, set this param to 1 (enable); otherwise set it to 0. (See note 2 below.)

velsen Velocity Sense enabled. If your synth responds to "Key Velocity" data in MIDI Note-On messages, you might want to enable this function, depending on which synth effect(s) you want to control with it. (See note 3 below.)

moden Modulation messages enabled. Set this param to 1 to allow sending MIDI modulation messages (Control Change #1). Set to 0 to disable.

pben Pitch Bend enabled. If your REMI has a Pitch Bend sensor fitted and your synth supports MIDI Pitch Bend messages, set this param to 1; otherwise set it to 0. (Planned future upgrade of REMI handset.)

The second group of parameters (below) defines hardware characteristics of your REMI including fingering scheme, pitch offset and sensor calibration data. There is only one set of parameters in this group. The Mode Switch has no effect on these settings.

fingsc Fingering Scheme. REMI supports four options (0, 1, 2 or 3) for the fingering scheme. This setting determines the mapping of finger/pad combinations to respective musical notes (pitch). Fingering schemes are dependent on the touch-pad layout. Options 0 and 2 assume a pad located at RH5. Options 1 and 3 assume a pad located at LH4. See section: "Fingering Schemes" for more details.

pitch Pitch Offset. This parameter sets the base musical note for the instrument in semitones. A pitch offset of zero (0) sets the base note to C4 (Middle C). The pitch offset value range is –24 to +24.

thres Touch Sense Threshold. This parameter sets the logic level threshold of the touch-pad sensing function. It is normally in the range 100 to 200. (Default = 150.) See also section: "Hardware Testing ..."

prspan Pressure Sensor Span. This parameter sets the sensor ADC reading span. See section: "Hardware Testing and Sensor Calibration".

modmax Modulation sensor maximum ADC reading (full-scale). See section: "Hardware Testing and Sensor Calibration".

modband Modulation sensor dead-band adjustment. ADC reading at upper end of dead-band range. Typically about 25% to 50% of modmax for FSR-402. See section: "Hardware Testing and Sensor Calibration".

Notes:

(1) Some sound modules and software synths are designed specifically for EWIs and respond to Breath Pressure Control Change commands (CC number = 2). If using this type of synth, set the MIDI Expression CC number = 2. Some MIDI synths respond only to "Channel Volume" messages (CC number = 7). If using such a device, set the MIDI Pressure CC number = 7.

(2) The REMI sends a MIDI Note-On/Velocity command when the breath pressure exceeds a preset threshold. A corresponding Note-Off command is sent when the breath pressure drops below the "note-off pressure level". With legato mode enabled, while a note is playing, a change in fingering will cause another Note-On/Velocity command to be transmitted without first sending a Note-Off command. If your sound module or synth is monophonic, or set to Mono mode, the change in fingering should result in a different note sounding, i.e. a change in pitch, without "re-triggering" the envelope generator(s). Some polyphonic synthesizers do not support MIDI legato mode and may not function properly with legato mode enabled. To disable it, set the REMI legato parameter to 0.

(3) Many touch-sensitive keyboard synthesizers respond to "key velocity" data received in MIDI Note-On/Velocity messages. The velocity data may be used to control audio amplitude (loudness) or various other synth effects, e.g. envelope attack rate, filter cut-off frequency, resonance, etc. For EWI controllers, it is generally preferred not to use velocity data to control audio amplitude, except if this is the only method supported by your synth. If you are not using velocity data to control a synth effect, it is best to disable it, because velocity sensing causes a delay in note triggering.

Pitch Offset Setting

A configuration parameter "Pitch Offset" is provided so that the range of musical notes produced by the REMI can be shifted, i.e. transposed, up or down by a number of semitones.

The parameter value may be set by a CLI command (**config pitch =** ##) to any value in the range −24 to +24 (semitones) or it may be set to one of 5 "preset" values using the PRESET button in conjunction with the octave pads (OCT-, OCT+) and pads at LH1 and LH3.

This feature allows the REMI to be easily configured to emulate recorders of different types, e.g. bass in C3, bass in F3, tenor in C4 (default), alto/treble in F5 and soprano/descant in C5.

To set one of the 5 pitch options, press the PRESET button while touching the pad(s) shown in the table below, while no note is being played.

	Bass C3	Bass F3	Tenor C4	Alto / Treble F4	Soprano / Descant C5
Pitch Offset	−12	−7	0	+5	+12
Touch-Pad(s)	OCT-	OCT- & LH3	OCT- & OCT+	OCT+ & LH1	OCT+

4. Instrument Presets

REMI 2 provides a facility to select 1 of 8 presets directly from the handset, without needing to use the synth control panel. The desired preset is selected by a button on the handset, in combination with the 8 upper touch-pads. MIDI Program Numbers corresponding to the 8 REMI presets can be set up using a CLI command.

To select a Preset: While holding the touch-pad corresponding to the preset required, press the PRESET button. Preset #1 corresponds to the lowest touch-pad (near the connector end). Preset #8 is the highest pad (nearest to the mouth-piece).

To allow the REMI to send MIDI Program Change messages, the configuration parameter "**prchen**" must be set to 1. If it is not already set to 1, enter the command:

```
config prchen = 1
```

Of course, you can always use the synth control panel to change the instrument voice. If you prefer this option, it is probably best to disable MIDI Program Change messages, so that the REMI will not send this message if the PRESET button is hit accidentally. To configure this option, simply enter the command:

```
config prchen = 0
```

As stated before, there is a MIDI program number assigned to each of the 8 REMI presets. To see these numbers, enter the command "**preset**" as follows:

```
> preset

_____________________
Preset   | MIDI Prgm #
`````````|`````````````
    1 | 75
    2 | 72
    3 | 69
    4 | 74
    5 | 17
    6 | 20
    7 | 23
    8 | 67
_________|_____________
```

To change any of the assigned program numbers, enter the command:

`preset` *p* `=` *nn*

... where *p* is a preset number (1 to 8) and *nn* is the desired MIDI program (voice) number.

### Note on ***Bauer**** EWI Synth Modules

If MIDI System Exclusive messages are enabled, i.e. if the parameter **sysxen** is set to true (1), then the assigned Program Numbers are not sent in MIDI Program Change messages. Instead, the Preset number (1 to 8) of the currently selected Preset will be sent in Program Change messages and in Sys-Ex 'PRESET' messages. This message protocol is compatible with both the REMI mk2 synth and Sigma-6 synth modules.

The REMI mk2 synth gets the Preset number from the Sys-Ex 'PRESET' message; Program Change messages are ignored. The Sigma-6 synth gets the Preset number from Program Change messages; Sys-Ex 'PRESET' messages are ignored.

| | REMI mk2 Synth | Sigma-6 Synth | Other Synth |
|---|---|---|---|
| Sys-Ex msg | PRESET # | IDENT only | N/A |
| Prgm Chg msg | ignored | Preset # | Program # |

The REMI mk2 synth and Sigma-6 synth modules both use the Bauer System-Exclusive 'IDENT' message type to detect connection of a REMI 2 handset.
```

5. Fingering Schemes

The **REMI "Standard" fingering scheme (0)** was designed to be easy to learn and to provide a large range of notes – up to four octaves. The player needs to learn only the finger positions of 12 notes comprising one octave. This scheme is best for beginners learning to play an EWI.

REMI Fingering Chart – Standard (RH5 flattens)

LH1	●	●	●	●	●	●	●	●	●	●	●	●	○
LH2	●	●	●	●	●	●	●	●	●	●	○	○	●
LH3	●	●	●	●	●	●	●	●	○	○	◇	○	○
RH1	●	●	●	●	●	●	○	○	◇	○	◇	○	**x**
RH2	●	●	●	●	●	○	◇	○	◇	○	◇	○	**x**
RH3	●	●	●	○	○	**x**	◇	○	◇	○	◇	○	**x**
RH4	●	○	○	◇	○	x	◇	○	◇	○	◇	○	x
RH5	x	●	○	◇	○	x	◇	○	◇	○	◇	○	x
	C'	C#' Db'	**D'**	Eb' D#'	**E'**	**F'**	F#' Gb'	**G'**	Ab' G#'	**A'**	Bb' A#'	**B'**	**C"** (Alt)

LH1	○	○	○	○	○	○	○	○	○	○	○	○	○
LH2	●	●	●	●	●	●	●	●	○	○	○	○	○
LH3	●	●	●	●	●	●	●	●	●	●	●	●	●
RH1	●	●	●	●	●	●	○	○	●	●	●	●	○
RH2	●	●	●	●	●	○	◇	○	●	●	○	○	**x**
RH3	●	●	●	○	○	**x**	◇	○	◇	○	◇	○	**x**
RH4	●	○	○	◇	○	x	◇	○	◇	○	◇	○	x
RH5	x	●	○	◇	○	x	◇	○	◇	○	◇	○	x
	C"	C#" Db"	**D"**	Eb" D#"	**E"**	**F"**	F#" Gb"	**G"**	Ab" G#"	**A"**	Bb" A#"	**B"**	**C'"**

Legend: ● Pad touched ○ Pad not touched
◇ Any pad(s) in group **x** Don't care (No effect)

Two "octave pads" (OCT-, OCT+) on the underside of the instrument select one of 3 ranges of notes, each range spanning up to 2 octaves, as shown in the chart. When both octave pads are touched, the "middle" note range is selected (normally C4 to C6). By moving the thumb to the upper or lower pad, the range of notes is shifted up or down by one octave, respectively. The octave pads extend the overall range to four octaves (and more using "pitch offset").

To select the "standard" scheme, set the configuration parameter "**fingsc**" = **0**.

REMI Alternative Fingering Scheme (1)

An alternative fingering scheme has a pad located at LH4 instead of RH5. This option uses pad LH4 to sharpen certain notes, i.e. C and F. The scheme was implemented in the REMI for compatibility with some commercial EWI's (e.g. the "Sylphyo") and a few other DIY designs which provide "recorder-like" fingering.

REMI Fingering Chart – Alternate (LH4 sharpens)

LH1	●	●	●	●	●	●	●	●	●	●	●	●	●	○
LH2	●	●	●	●	●	●	●	●	●	●	●	○	○	●
LH3	●	●	●	●	●	●	●	●	●	○	○	◇	○	○
LH4	○	●	x	x	x	○	●	x	x	x	x	x	x	○
RH1	●	●	●	●	●	●	●	○	○	◇	○	◇	○	x
RH2	●	●	●	●	●	○	○	◇	○	◇	○	◇	○	x
RH3	●	●	●	○	○	x	x	◇	○	◇	○	◇	○	x
RH4	●	●	○	◇	○	x	x	◇	○	◇	○	◇	○	x
	C'	C#' Db'	**D'**	Eb' D#'	**E'**	**F'**	F#' Alt.	F#' Gb'	**G'**	Ab' G#'	**A'**	Bb' A#'	**B'**	**C"**

LH1	○	○	○	○	○	○	○	○	○	○	○	○	○	○
LH2	●	●	●	●	●	●	●	●	●	○	○	○	○	○
LH3	●	●	●	●	●	●	●	●	●	●	●	●	●	●
LH4	○	●	x	x	x	○	●	x	x	x	x	x	x	○
RH1	●	●	●	●	●	●	●	○	○	●	●	●	●	○
RH2	●	●	●	●	●	○	○	◇	○	●	●	○	○	x
RH3	●	●	●	○	○	x	x	◇	○	◇	○	◇	○	x
RH4	●	●	○	◇	○	x	x	◇	○	◇	○	◇	○	x
	C"	C#" Db"	**D"**	Eb" D#"	**E"**	**F"**	F#" Alt.	F#" Gb"	**G"**	Ab" G#"	**A"**	Bb" A#"	**B"**	**C'"**

Legend: ● Pad touched ○ Pad not touched
◇ Any pad(s) in group x Don't care (No effect)

The octave pads operate in the same manner as in the "standard" fingering scheme.

To select the "alternative" scheme, set the configuration parameter "**fingsc**" = **1**.

Baroque Fingering Scheme (2)

The Baroque fingering scheme provides realistic emulation of modern acoustic recorders, both German and English, when played on the REMI. Some of the German fingerings may sound out of tune when played on an English acoustic recorder, because there are physical differences in the German and English recorder designs. Electronic instruments do not suffer from such physical design constraints, and so compatibility is achieved.

There are some small compromises in the REMI implementation of the Baroque fingerings. Acoustic recorders require a playing technique called "partial hole covering" (*aka* "half-holing") in the thumb hole and in the position of RH4 (right-hand little finger). Most modern recorders have two small holes in this position, where one or both may be covered.

The REMI equivalent of "full-hole" cover of the thumb hole is to touch the octave pad OCT+ (or both OCT+ and OCT- together). The "half-hole" equivalent is to touch only OCT-. The "half-hole" equivalent of RH4 is to touch pad RH5 instead of RH4 (not both together). Some fingering charts may indicate half-hole operation of RH3. This has no equivalent in the REMI, but the required note pitch should result regardless of whether RH3 is touched or not.

To select Baroque fingering, set the configuration parameter "**fingsc**" = **2**.

For REMI makers who prefer a physical layout with a pad located at LH4 instead of RH5, this option is allowed by setting "**fingsc**" = **3**. This does not change the operation of the extra pad. It will have the same effect, i.e. to emulate RH4 "half-hole", regardless of whether the pad is located at LH4 or RH5. (The micro-controller needs to know the physical pad layout so that the PRESET button functions properly.)

Most modern recorder fingering charts should be compatible with REMI scheme #2. Search the web for a suitable chart, for example: waterpigs.co.uk/recorder (Barnaby Walters)

To translate from a real recorder chart to REMI fingering, follow these simple rules:

- The REMI equivalent of thumb "half hole" pinch is to touch pad OCT- without OCT+.
- The REMI equivalent of RH4 (little finger) "half hole" is to touch pad RH5 without RH4.

Verification of Fingering

A REMI CLI facility is provided to check fingering. Enter the command "**diag -n 1**" to activate the "Note-On Display" function. Whenever a note is initiated, the note will be displayed in the terminal window. The standard musical notation (e.g. C4, A5, F#5) and MIDI note number will be shown. This facility can be used to check that the correct (or expected) note is produced for any given finger combination.

Kindly notify the author if you suspect that a wrong note is produced by a finger pattern found in a recorder chart, or if there is no finger pattern available to produce a desired note. Be sure to send me a copy of the chart in question.

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